

Vol 11 Chapter 10 — Mersenne Primes and Lucas-Lehmer

Keeper (author pass — deep math/physics revision)

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Chapter 10 — Mersenne Primes and Lucas-Lehmer

Level 1 — one sentence

Mersenne primes $M_p = 2^p - 1$ for prime p have been studied since antiquity, with Lucas-Lehmer primality test providing efficient verification, and the BST team's Friday May 22, 2026 finding ("Mersenne Network Convergence") shows BST primary exponents are anchored in a Mersenne ladder: $M_{\text{rank}} = 3 = N_c$, $M_{N_c} = 7 = g$, $M_{n_c} = 31$, $M_g = 127$ — with additive identity $N_{\text{max}} - M_g = 10 = g + N_c$.

Level 2 — graduate-physicist precision

10.1 Mersenne primes

$M_p = 2^p - 1$. Prime M_p requires p prime, but not all prime p give prime M_p .

Known Mersenne primes (52 as of 2024): $M_2 = 3, M_3 = 7, M_5 = 31, M_7 = 127, M_{13} = 8191, M_{17} = 131071, M_{19}, M_{31}, M_{61}, M_{89}, M_{107}, M_{127}, M_{521}, \dots, M_{82589933}$.

GIMPS (Great Internet Mersenne Prime Search) project actively searches.

10.2 Lucas-Lehmer primality test

For $p \geq 3$ odd prime: define $s_0 = 4$ and $s_{i+1} = s_i^2 - 2 \pmod{M_p}$.

M_p is prime iff $s_{p-2} \equiv 0 \pmod{M_p}$.

Polynomial time in p — efficient.

10.3 BST Mersenne ladder (Friday May 22, 2026)

The team's Friday May 22, 2026 substrate-investigation finding (Mersenne Network Convergence):

BST primary exponents are anchored on Mersenne primes: - $M_{\text{rank}} = M_2 = 3 = N_c$ (Mersenne exponent rank $\rightarrow$ BST primary N_c) - $M_{N_c} = M_3 = 7 = g$ (Mersenne exponent $N_c \rightarrow$ BST primary g) - $M_{n_c} = M_5 = 31$ (Mersenne exponent $n_c \rightarrow 31$, prime) - $M_g = M_7 = 127$ (Mersenne exponent $g \rightarrow 127$, prime)

Additive identity: $N_{\text{max}} - M_g = 137 - 127 = 10 = g + N_c$. The substrate's natural integer relationship.

The $c_2 = 6$ "gap" resolved via $M_{c_2} = M_6 = 63 = 9 \cdot 7$, requiring composite interpretation.

10.4 6/7 Mersenne primes

6 of BST's first 7 primary-exponent Mersenne values are themselves prime: $\{3, 7, 31, 127\}$ are Mersenne primes. Only $M_6 = 63$ is composite (as expected — index 6 is composite).

This high "Mersenne primality density" in BST primary exponents is a substrate-mechanism finding (Lyra T2452, Friday May 22, 2026).

10.5 Substrate Mersenne network

Combined with Graph Forces (Grace) and Mersenne tower (Elie + Casey): three independent patterns converge on substrate Mersenne-anchored over-determination of BST primaries: - Graph Forces (Grace, $p \approx 2.7 \times 10^{-5}$) - Mersenne tower (T2452 Elie+Casey) - Mersenne ladder (Friday May 22 finding)

Triple convergence is one of BST's strong-uniqueness pieces of evidence.

10.6 Substrate computational substrate

The substrate's natural RS coding on $\text{GF}(2^8) = \text{GF}(128)$ (Ch 9) has maximum codeword length $n = 127 = M_g = M_7$.

Substrate has a Mersenne prime as its natural codeword-length parameter — direct substrate-mechanism integer.

10.7 K-audit anchors

- Mersenne ladder Friday May 22, 2026
- Lyra T2452: Mersenne tower theorem
- K156 audit anchor (pending): three-layer over-determinism with Mersenne layer

Level 3 — 5th-grader accessibility

Mersenne primes: $M_p = 2^p - 1$ for prime p . Examples: $M_2 = 3, M_3 = 7, M_5 = 31, M_7 = 127$. GIMPS project hunts for more. BST Mersenne ladder (May 22, 2026 finding): the BST primary integers fit into a Mersenne pattern: $M_2 = N_c, M_3 = g$, etc. 6 of 7 are Mersenne primes. Additive identity: $N_{\max} - M_g = 10 = g + N_c$. The substrate's natural integer arithmetic is Mersenne-anchored. Codeword length $127 = M_7 = M_g$ for substrate's Reed-Solomon coding.

What comes next

Chapter 11 develops the generative-geometry framing thesis.

Where to look this up

- GIMPS project
- Crandall-Pomerance, *Prime Numbers*
- BST: Mersenne ladder Friday May 22; Lyra T2452